

PATENT RESEARCH REPORT

1. ABSTRACT

This comprehensive technical report synthesizes the polymerization, metathesis, and formulation landscape of Hydrogenated Nitrile Butadiene Rubber (HNBR) based on fifteen key patents. The research scope covers the transition from standard high-viscosity acrylonitrile-butadiene rubber (NBR) to low molecular weight, highly processable HNBR. The selected patent landscape reveals two primary polymerization and modification approaches: (1) a two-step process involving metathetic degradation of NBR in the presence of a co-olefin (such as 1-hexene or ethylene) followed by catalytic hydrogenation, and (2) a single-step, simultaneous metathesis and hydrogenation process. Key formulation and process trends highlight the use of highly active ruthenium-based carbene catalysts (e.g., Grubbs, Hoveyda, and Grela catalysts) to control molecular weight and polydispersity. Additionally, the report covers post-hydrogenation treatments, such as the oxidation of residual phosphine cocatalysts to phosphine oxides to improve vulcanizate properties, and the optimization of residual catalyst concentrations to enhance thermal oxidative aging without requiring complex purification steps. Finally, industrial applications of HNBR in high-temperature pipeline insulation and additive manufacturing (3D printing) are detailed, demonstrating the polymer's versatile performance profile. The report provides a deep dive into the reaction parameters, including monomer ratios, catalyst loadings, temperatures, pressures, and reaction times, offering a valuable reference for polymer scientists and research analysts working on advanced elastomeric materials. By analyzing these diverse patent disclosures, this report establishes a clear correlation between synthesis conditions and final polymer properties, guiding the development of next-generation HNBR formulations.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP2473281B1
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: EP
- Publication Year: 2014
- Relevance: Describes a process for producing low molecular weight HNBR by simultaneous hydrogenation and metathesis of NBR in the presence of hydrogen and a ruthenium or osmium catalyst.

Polymerization / Synthesis Method - Monomer: 1,3-butadiene (conjugated diene) - Monomer: acrylonitrile (unsaturated nitrile) - Catalyst: Ruthenium or osmium complex of formula (I) - Co-olefin: Optional - Reaction Type: Simultaneous metathesis and hydrogenation

Relevant Experimental Evidence - The patent describes a simultaneous hydrogenation and metathesis process of nitrile rubber (NBR) in the presence of hydrogen and a ruthenium or osmium catalyst of formula (I). - The process yields hydrogenated nitrile rubber (HNBR) with a molecular weight (Mw) in the range of 20,000 to 250,000 g/mol. - The resulting HNBR has a Mooney viscosity (ML 1+4 @ 100°C) in the range of 1 to 50 and a narrow molecular weight distribution (polydispersity index).

Technical Relevance Provides a single-step method for simultaneous metathesis (molecular weight reduction) and hydrogenation of NBR to produce low-viscosity HNBR without requiring a separate degradation step.

Patent 2

- Patent Number: US9079979B2
- Patent Title: Process for the preparation of hydrogenated nitrile rubber
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 2015
- Relevance: Focuses on simultaneous hydrogenation and metathesis of NBR using specific ruthenium or osmium catalysts (e.g., Grubbs III, Grubbs II, or Wilkinson's catalyst) to obtain low molecular weight HNBR.

Polymerization / Synthesis Method - Catalyst: Grubbs III Catalyst, Grubbs II Catalyst, or Wilkinson's Catalyst - Catalyst Amount: 0.057 phr (HNBR 1), 0.080 phr (HNBR 2), 0.06 phr (HNBR 3), 0.055 phr (HNBR 4) - Reaction Time: 120 min - Monomers: 1,3-butadiene and acrylonitrile

Relevant Experimental Evidence - Example 1 (Table 1) compares hydrogenation parameters for HNBR 1, 2, 3, and 4 using different catalysts and amounts. - HNBR 1 and 2 utilize Grubbs III catalyst at 0.057 phr and 0.080 phr respectively, while HNBR 3 uses Wilkinson's catalyst (0.06 phr) and HNBR 4 uses Grubbs II catalyst (0.055 phr). - The reaction is carried out for 120 minutes to achieve simultaneous metathesis and hydrogenation, resulting in low molecular weight HNBR with Mw of 20,000 to 250,000 g/mol and Mooney viscosity of 1 to 50.

Technical Relevance Demonstrates the efficiency of Grubbs-type catalysts in performing simultaneous metathesis and hydrogenation, showing improved hydrogenation efficiency and metathesis activity.

Patent 3

- Patent Number: US20210340285A1
- Patent Title: Method for producing hydrogenated nitrile rubber and hnbr compositions thereof

- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 2021
- Relevance: Discloses a method for producing HNBR with good aging properties by controlling the residual amount of noble metal catalysts (Ru, Pd, Rh) without requiring complex catalyst removal steps.

Polymerization / Synthesis Method - Monomer Content: 40% to 90% by weight of a conjugated diene, 10% to 60% by weight of an alpha,beta-unsaturated nitrile - Catalyst: Ruthenium compound, Palladium compound, or Rhodium compound - Ruthenium Concentration: 10 ppm to 200 ppm (preferably 42 ppm to 79 ppm) based on NBR - Palladium Concentration: 20 ppm to <67 ppm (or 44 ppm to <67 ppm) based on NBR - Rhodium Concentration: 50 ppm to <270 ppm (preferably 79 ppm to 170 ppm) based on NBR - Temperature: 60 to 200 °C - Pressure: 700,000 Pa to 15,000,000 Pa (7 to 150 bar) - Reaction Time: 1 to 24 hours

Relevant Experimental Evidence - The method subjects NBR in solution to hydrogenation using controlled low amounts of Ru, Pd, or Rh catalysts. - Specific ruthenium catalysts include RuHCl(CO)(PPh₃)₃, Grubbs first-generation, and Grubbs second-generation catalysts. - The resulting HNBR exhibits improved aging properties: gel content increases by less than 10% and Mooney viscosity increases by less than 40% after aging for 4 days at 140°C.

Technical Relevance Eliminates the need for expensive and complex catalyst removal steps by optimizing the catalyst concentration to a level that does not detrimentally affect the thermal oxidative aging of the HNBR.

Patent 4

- Patent Number: US7919563B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 2011
- Relevance: Details a two-step or one-pot process for preparing low molecular weight HNBR with narrow polydispersity using metathesis followed by hydrogenation.

Polymerization / Synthesis Method - Rubber Concentration in MCB: 15 wt% (Examples 1, 2) or 6 wt% (Examples 3, 4) - Rubber Amount: 200 g (Examples 1, 2) or 75 g (Examples 3, 4) - Monochlorobenzene (MCB) Solvent: 1133 g (Examples 1, 2) or 1175 g (Examples 3, 4) - Co-olefin: Ethylene (C₂H₄) or 1-hexene - Co-olefin Amount: 150 g (1-hexene in Example 4) or 500 psi (ethylene in Examples 1-3) - Catalyst Load: 0.05 phr, 0.10 phr, or 0.25 phr - Reactor Temperature: 80 °C

Relevant Experimental Evidence - In Examples 1 and 2, 200 g of NBR was dissolved in 1133 g of MCB (15 wt% solid) and reacted with ethylene co-olefin at 500 psi. - In Example 3, 75 g of NBR

was dissolved in 1175 g of MCB (6 wt% solid) with ethylene at 500 psi and a catalyst load of 0.05 to 0.25 phr. - In Example 4, 75 g of NBR was dissolved in 1175 g of MCB (6 wt% solid) and 150 g of 1-hexene was added as co-olefin. - The process successfully reduces the molecular weight of NBR (e.g., Mn from 85 kg/mol to 27 kg/mol, Mw from 296 kg/mol to 54 kg/mol) to yield low-viscosity HNBR with a polydispersity index of less than 2.5.

Technical Relevance Provides detailed reaction conditions for metathesis degradation of NBR using co-olefins (ethylene or 1-hexene) to control the final molecular weight and viscosity of the resulting HNBR.

Patent 5

- Patent Number: US20160326272A1
- Patent Title: Hydrogenated nitrile rubber containing phosphine oxide or diphosphine oxide
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 2016
- Relevance: Discloses HNBR compositions containing specific amounts of phosphine oxides and halogens, which are treated with oxidizing agents to improve vulcanizate properties like modulus and compression set.

Polymerization / Synthesis Method - Phosphine/Diphosphine Content: 0 to 1.0 wt% - Phosphine Oxide/Diphosphine Oxide Content: 0.25 to 6 wt% - Total Halogen Content: 25 to 10000 ppm - Triphenylphosphine (TPP) Cocatalyst: 2.9 parts by weight per 100 g of NBR (11.06 mmol/100 g NBR) - Oxidizing Agent: Potassium chlorate, potassium perchlorate, potassium permanganate, potassium dichromate, or sodium hypochlorite - Sodium Hydroxide: 1370 g in 6150 g deionized water (for sodium hypochlorite preparation) - Chlorine Gas: 1200 g introduced at 5-10 °C - Vulcanization Pressure: 120 bar

Relevant Experimental Evidence - In Experiment Series 5, HNBR produced using 2.9 parts by weight of TPP per 100 g of NBR was reacted with oxidizing agents in the solid state. - Sodium hypochlorite was prepared by dissolving 1370 g of NaOH in 6150 g of deionized water, introducing 1200 g of chlorine gas at 5-10°C, and adjusting the pH to >12.5 with 50 g of 48% NaOH solution. - The oxidizing agents convert residual phosphines (e.g., TPP) to phosphine oxides (e.g., TPPO), resulting in vulcanizates with improved modulus (e.g., stress values at 10%, 25%, 50%, 100%, 200%, and 300% strain) and lower compression set.

Technical Relevance Addresses the negative impact of residual phosphine cocatalysts (used during homogeneous hydrogenation) on vulcanization by converting them into beneficial phosphine oxides via controlled oxidation.

Patent 6

- Patent Number: US9023914B2
- Patent Title: Nitrile rubbers and production thereof in organic solvents

- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 2015
- Relevance: Describes a process for preparing NBR by free-radical polymerization in an organic solvent using RAFT modifiers, which can be subsequently hydrogenated to HNBR.

Polymerization / Synthesis Method - Monomer: 1,3-butadiene (62 phm, 71 ml, 847 mmol) - Monomer: Acrylonitrile (34 ml or 35.6 g) - Initiator: Vazo 88 (128 mg, 0.15 phm) in 5 ml dimethylacetamide (DMAc) - RAFT Modifier: Dibenzoyl Trithiocarbonate (DiBenT) or Cumyl Phenyl Dithioacetate (CPDA) - Solvent: Dimethylacetamide (DMAc) (2 g used in Example 8) - Polymerization Temperature: 100 °C - Polymerization Pressure: 3 bar - Reaction Time: 6 hours, 13 hours, or 22 hours - Final Conversion: 9.2%, 39%, or 48% - Antioxidant: Vulkanox BKF (50% dispersion, 0.3 wt% based on NBR solids)

Relevant Experimental Evidence - Examples 9-16 describe the polymerization of acrylonitrile (34 ml) and 1,3-butadiene (71 ml) in a 5 L autoclave under 3 bar pressure at 100°C. - In Example 8, 2.74 g of NBR macro-RAFT agent ($M_n = 12,000$ g/mol) was dissolved in 2 g of DMAc and copolymerized with monomers. - In Examples 18, 20, and 23, Vazo 88 initiator (128 mg) in 5 ml of DMAc was added during polymerization. - The process achieves controlled polymerization with narrow polydispersity indices, and the resulting NBR can be directly hydrogenated in the same organic solvent to produce HNBR.

Technical Relevance Demonstrates a solution polymerization route for NBR using RAFT technology, which avoids the need to isolate NBR from an aqueous emulsion before conducting homogeneous hydrogenation to HNBR.

Patent 7

- Patent Number: EP2519350B1
- Patent Title: Highly active metathesis catalysis selective for romp and rcm
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: EP
- Publication Year: 2019
- Relevance: Discloses highly active ruthenium carbene complexes with six-coordinate bonds containing a Ru-N bond, useful for NBR depolymerization and subsequent hydrogenation to HNBR.

Polymerization / Synthesis Method - NBR Substrate: 60 g - Solvent: Anhydrous chlorobenzene (500 mL) - Reactor: 1.0 L steel well-sealed reactor - Catalyst Load: 0.07 wt% or 0.1 wt% Ru complex - Reaction Temperature: 130 °C (overnight) - Hydrogen Pressure: 5 MPa - Product Yield: 98%

Relevant Experimental Evidence - In the metathesis and hydrogenation examples, 60 g of NBR substrate was dissolved in 500 mL of anhydrous chlorobenzene in a 1.0 L steel reactor. - Ru

complex catalyst (0.07 wt% or 0.1 wt%) was added, and the mixture was reacted under 5 MPa of hydrogen pressure at 130°C overnight. - The depolymerized and hydrogenated rubber product was precipitated in methanol, yielding HNBR at over 98% yield.

Technical Relevance Introduces novel, highly active, and stable ruthenium catalysts that can perform both NBR depolymerization and high-pressure hydrogenation in a single system.

Patent 8

- Patent Number: EP2292687B1
- Patent Title: Improved vulcanisable guanidine-free mixtures containing ethylene acrylate (AEM) ...
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: EP
- Publication Year: 2012
- Relevance: Describes guanidine-free vulcanizable mixtures containing HNBR-based rubbers with carboxylic acid groups (terpolymers) and diamine crosslinking systems.

Polymerization / Synthesis Method - NBR 1 Composition: 60.4 wt% butadiene, maleic acid monoethyl ester (termonomer), 39.6 wt% acrylonitrile - Mooney Viscosity: 44, 45, 47, or 48 MU (ML 1+4 @ 100°C) - Hydrogenation Catalyst: Wilkinson's catalyst with triphenylphosphine (TPP) co-catalyst - Hydrogenation Pressure: 85 bar - Hydrogenation Temperature: 138 °C - Reaction Time: 4 hours - Cooling Temperature: ~22 °C - Vulcanization Temperature: 180 °C

Relevant Experimental Evidence - HNBR 1 was prepared by hydrogenating a terpolymer (NBR 1) containing 60.4 wt% butadiene, maleic acid monoethyl ester, and acrylonitrile using Wilkinson's catalyst and TPP at 138°C and 85 bar for 4 hours. - The resulting HNBR 1 had a residual double bond content of 0.4% and a maleic monoethylester content of 5%. - Guanidine-free vulcanizable mixtures were prepared using HNBR 1, hexamethylenediamine carbamate (Rhenogran HMDC-70/AEMD), and retarders like benzoic acid, showing high scorch safety at 120°C and excellent curing profiles at 180°C.

Technical Relevance Focuses on the synthesis and compounding of carboxylated HNBR (HXNBR) terpolymers, which are crosslinked using diamine systems to achieve high heat and compression set resistance.

Patent 9

- Patent Number: EP2473278B1
- Patent Title: Metathesis of nitrile rubbers in the presence of transition metal catalysts
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: EP
- Publication Year: 2014

- **Relevance:** Discloses a process for preparing low molecular weight HNBR ($M_w \leq 50,000$ g/mol, $PDI < 2.0$) by metathesis of NBR followed by isolation using a mechanical degassing device.

Polymerization / Synthesis Method - NBR Rubber: Perbunan T 3435 (700 g) - Solvent: Monochlorobenzene (MCB) (4667 g) - Reactor: 10 L high pressure reactor - Co-olefin: 1-hexene (4 phr) - Metathesis Temperature: 22 °C (for 2 hours) - Hydrogenation Temperature: 138 °C (for 4 hours) - Agitation Speed: 600 rpm

Relevant Experimental Evidence - In Examples 5-6, 700 g of Perbunan T 3435 was dissolved in 4667 g of MCB in a 10 L reactor. - 4 phr of 1-hexene was added and agitated for 2 hours at 22°C, followed by addition of a ruthenium metathesis catalyst. - The reaction mixture was then hydrogenated at 138°C for 4 hours under agitation (600 rpm) to obtain low molecular weight HNBR. - The polymer was isolated from the solvent using a mechanical degassing device to handle the extreme tackiness of the low molecular weight rubber.

Technical Relevance Solves the isolation problem of highly tacky, low molecular weight HNBR by utilizing a mechanical degassing device instead of conventional steam coagulation.

Patent 10

- Patent Number: US7572861B2
- Patent Title: HXNBR-rubber as a cross-linking agent
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 2009
- Relevance: Focuses on HXNBR containing molecularly dispersed triphenylphosphine (TPP) as a crosslinking and elasticizing agent for other elastomers and plastics.

Polymerization / Synthesis Method - TPP Dispersion Scale: No difference in concentration on a dimensional scale greater than 1 micrometer - TPP Concentration: 0.1 to 10 wt% in HXNBR - Double Bond Content: Less than 40 double bonds per 1000 carbon atoms (preferably 0.2 to 15) - Solvents: Chlorobenzene, acetone, methyl ethyl ketone, methylene chloride, chloroform, tetrahydrofuran, or toluene

Relevant Experimental Evidence - The patent describes preparing HXNBR by emulsion polymerization of butadiene, acrylonitrile, and carboxylated monomers (e.g., acrylic acid, methacrylic acid) to form XNBR, followed by homogeneous hydrogenation. - The homogeneous hydrogenation process directly equips the HXNBR with molecularly dispersed TPP (0.1 to 10 wt%) as a residual catalyst/cocatalyst component. - This molecularly dispersed TPP acts as an in-situ activator for subsequent crosslinking reactions with epoxides, isocyanates, or amines when blended with other polymers.

Technical Relevance Utilizes the residual TPP from the homogeneous hydrogenation of XNBR as a molecularly dispersed activator, turning a typical impurity into a functional crosslinking

promoter.

Patent 11

- Patent Number: EP2057205B1
- Patent Title: Production of meta-block copolymers by polymer segment interchange
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: EP
- Publication Year: 2017
- Relevance: Describes the preparation of meta-block copolymers by contacting a metathesis catalyst with a mixture of polymer reagents, including HNBR.

Polymerization / Synthesis Method - Polymer Reagents: Partially-hydrogenated polybutadiene (HPBD) (0.25 g) and partially-hydrogenated nitrile/butadiene rubber (HNBR) (0.25 g) - Solvent: Toluene - Catalyst: Grubbs II metathesis catalyst (8 mg or 9 mg) - Reaction Temperature: 95 °C or 105 °C - Extraction Solvent: Tetrahydrofuran (30 ml or 50 ml)

Relevant Experimental Evidence - In Example 1, a toluene solution containing 0.25 g of HPBD and 0.25 g of HNBR was warmed to 95°C, and 8 mg of Grubbs II catalyst was added. - Volatile components were removed under reduced pressure to give 0.45 g of recovered meta-block copolymer product. - The resulting block copolymer was analyzed by SAXS and showed microphase order in the liquid phase at temperatures up to 300°C, confirming successful segment interchange.

Technical Relevance Demonstrates that HNBR can undergo metathetic segment interchange with other unsaturated polymers to form novel block copolymers with unique thermoplastic elastomeric properties.

Patent 12

- Patent Number: EP3030824B1
- Patent Title: Method for the production of and high temperature field joints
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: EP
- Publication Year: 2024
- Relevance: Utilizes HNBR as a high-temperature resistant polymer composition for pipeline insulation and field joints operating at temperatures up to 180°C or 205°C.

Polymerization / Synthesis Method - Continuous Operating Temperature: 150 °C to above 205 °C - HNBR Layer Maximum Operating Temperature: ~180 °C - Thermal Conductivity: Less than 0.40 W/mK (typically 0.10 to 0.35 W/mK) - Preheating Temperature: 100 °C - Thickness of Coating: 1000 micrometers

Relevant Experimental Evidence - The patent describes a pipeline insulation system where the innermost layer (22A) comprises HNBR, providing a maximum operating temperature of about 180°C. - The pipe sections are preheated to 100°C prior to spraying a chamfer inhibition barrier to a thickness of 1000 micrometers. - The HNBR composition provides excellent thermal insulation (thermal conductivity < 0.40 W/mK) and long-term heat resistance under continuous high-temperature operation.

Technical Relevance Highlights a high-value industrial application of HNBR in offshore oil and gas pipelines, leveraging its exceptional thermal stability, chemical resistance, and mechanical toughness.

Patent 13

- Patent Number: US11192293B2
- Patent Title: Process for producing 3D structures from rubber material
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 2021
- Relevance: Discloses an additive manufacturing (3D printing) process using high-viscosity rubber materials, specifically mentioning HNBR (Therban AT 3404) and NBR (Perbunan 2831 F).

Polymerization / Synthesis Method - HNBR Polymer: Therban AT 3404 (100 phr) - NBR Polymer: Perbunan 2831 F (100 phr) - Mooney Viscosity before curing: >10 MU at 60 °C and <200 MU at 100 °C - Nozzle Temperature: 100 °C - Preheating Time: 2 minutes - Curing Temperature: 180 °C - Curing Time: 15 minutes (or t80 + 5 min) - Curing Pressure: 10 bar water vapor, 5 bar under inert liquid, or atmospheric pressure - Carbon Black Filler: Corax N 550/30 (30 phr)

Relevant Experimental Evidence - The examples utilize Therban AT 3404 (HNBR) or Perbunan 2831 F (NBR) at 100 phr compounded with 30 phr of Corax N 550/30 carbon black. - The compound is extruded through a nozzle at 100°C after 2 minutes of preheating to form a 3D precursor. - The precursor is cured at 180°C for 15 minutes under atmospheric pressure or up to 10 bar water vapor pressure, yielding a shape-stable rubber article with Shore A hardness of 22 to 50.

Technical Relevance Establishes the rheological window (Mooney viscosity limits) required to successfully 3D print HNBR and NBR compounds without phase transition before curing.

Patent 14

- Patent Number: US8247063B2
- Patent Title: Multilayer material and related methods
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US

- Publication Year: 2012
- Relevance: Mentions the use of nitrile-based copolymers and elastomeric polyolefins in multilayer packaging materials.

Polymerization / Synthesis Method - Stretching Temperature: 90 to 140 °C (or 100-135 °C, 105-130 °C) - Draw Ratio: 1.1:1 to 9:1 (typically 4.6:1 to 5.4:1 or 5.5:1 to 6.5:1) - Adhesive Coating Weight: 10 to 50 gsm - Line Speed: 150 fpm

Relevant Experimental Evidence - No specific experimental examples of HNBR synthesis are disclosed in the supplied evidence.

Technical Relevance Provides context on the incorporation of elastomeric and nitrile-containing polymer layers in coextruded multilayer films.

Patent 15

- Patent Number: US9593168B2
- Patent Title: Process for degrading nitrile rubbers in the presence of catalysts having an ...
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 2017
- Relevance: Focuses on the metathetic degradation of NBR using highly active catalysts (e.g., Grela catalyst, Hoveyda catalyst) to prepare low-viscosity feedstocks for HNBR.

Polymerization / Synthesis Method - Catalyst: Grela catalyst, Hoveyda catalyst, or Grubbs II catalyst - Ruthenium Concentration: 8.6 ppm - Co-olefin: 1-hexene - Co-olefin Amount: 2.0 phr - Reaction Time: 185 minutes (Grela catalyst) or >425 minutes (Grubbs II)

Relevant Experimental Evidence - Example Series 4 and 5 compare the activity of the Grela catalyst with the Grubbs II catalyst in the presence of 2 phr of 1-hexene. - Using the Grela catalyst, final values for Mw and Mn were achieved after about 185 minutes, whereas the degradation was incomplete after 425 minutes using the Grubbs II catalyst. - In another example, NBR was degraded using 8.6 ppm of ruthenium from a Hoveyda catalyst and 2.0 phr of 1-hexene co-olefin.

Technical Relevance Demonstrates that using advanced ruthenium catalysts with specific ligand structures (like the Grela or Hoveyda catalysts) significantly accelerates the metathesis degradation of NBR, optimizing the feedstock preparation for HNBR.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- Monomer Content and Ratios: US20210340285A1 specifies NBR containing 40% to 90% by weight of conjugated diene and 10% to 60% by weight of unsaturated nitrile. EP2292687B1

utilizes a terpolymer (NBR 1) with 60.4 wt% butadiene, 39.6 wt% acrylonitrile, and a maleic acid monoethyl ester monomer.

- Polymerization and Degradation Processes: US9023914B2 describes the direct synthesis of NBR in organic solvents (DMAc) using RAFT modifiers, avoiding emulsion coagulation. In contrast, US7919563B2, EP2473278B1, and US9593168B2 focus on the metathetic degradation of pre-formed NBR in monochlorobenzene using co-olefins like 1-hexene or ethylene to reduce molecular weight prior to or during hydrogenation.
- Catalyst Systems: EP2473281B1 and US9079979B2 utilize ruthenium or osmium complexes for simultaneous metathesis and hydrogenation. US9593168B2 demonstrates that the Grela catalyst achieves complete NBR degradation in 185 minutes compared to over 425 minutes for the Grubbs II catalyst.
- Post-Hydrogenation and Additives: US20160326272A1 and US7572861B2 address residual triphenylphosphine (TPP) from homogeneous hydrogenation. US20160326272A1 uses oxidizing agents (e.g., sodium hypochlorite) to convert TPP to phosphine oxide to improve modulus, while US7572861B2 utilizes the molecularly dispersed TPP as an in-situ crosslinking activator.

4. CONCLUSION

In conclusion, the synthesis of high-performance Hydrogenated Nitrile Butadiene Rubber (HNBR) requires precise control over molecular weight, polydispersity, and residual catalyst chemistry. The evidence indicates that simultaneous metathesis and hydrogenation using advanced ruthenium carbene catalysts (such as Grubbs III or Grela-type complexes) offers a highly efficient, single-step route to low-viscosity HNBR (Mooney viscosity 1 to 50, Mw 20,000 to 250,000 g/mol). For applications requiring superior aging properties, maintaining residual noble metal concentrations within strict limits (e.g., Ru at 10–200 ppm, Pd at 20–67 ppm) avoids the need for costly catalyst removal while preventing gelation. Furthermore, converting residual phosphine cocatalysts to phosphine oxides via post-hydrogenation oxidation significantly enhances the modulus and compression set of the resulting vulcanizates. Recommended synthesis parameters include conducting reactions in monochlorobenzene at temperatures of 100–138°C, pressures of 5–8.5 MPa, and utilizing 2–4 phr of 1-hexene as a co-olefin for molecular weight control.

5. REFERENCES

- EP2473281B1 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473281B1/en>
- US9079979B2 | Process for the preparation of hydrogenated nitrile rubber | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9079979B2/en>
- US20210340285A1 | Method for producing hydrogenated nitrile rubber and hnbr compositions thereof | Not disclosed | US | 2021 | <https://patents.google.com/patent/US20210340285A1/en>

- US7919563B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7919563B2/en>
- US20160326272A1 | Hydrogenated nitrile rubber containing phosphine oxide or diphosphine oxide | Not disclosed | US | 2016 | <https://patents.google.com/patent/US20160326272A1/en>
- US9023914B2 | Nitrile rubbers and production thereof in organic solvents | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9023914B2/en>
- EP2519350B1 | Highly active metathesis catalysis selective for romp and rcm | Not disclosed | EP | 2019 | <https://patents.google.com/patent/EP2519350B1/en>
- EP2292687B1 | Improved vulcanisable guanidine-free mixtures containing ethylene acrylate (AEM ... | Not disclosed | EP | 2012 | <https://patents.google.com/patent/EP2292687B1/en>
- EP2473278B1 | Metathesis of nitrile rubbers in the presence of transition metal catalysts | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473278B1/en>
- US7572861B2 | HXNBR-rubber as a cross-linking agent | Not disclosed | US | 2009 | <https://patents.google.com/patent/US7572861B2/en>
- EP2057205B1 | Production of meta-block copolymers by polymer segment interchange | Not disclosed | EP | 2017 | <https://patents.google.com/patent/EP2057205B1/en>
- EP3030824B1 | Method for the production of and high temperature field joints | Not disclosed | EP | 2024 | <https://patents.google.com/patent/EP3030824B1/en>
- US11192293B2 | Process for producing 3D structures from rubber material | Not disclosed | US | 2021 | <https://patents.google.com/patent/US11192293B2/en>
- US8247063B2 | Multilayer material and related methods | Not disclosed | US | 2012 | <https://patents.google.com/patent/US8247063B2/en>
- US9593168B2 | Process for degrading nitrile rubbers in the presence of catalysts having an ... | Not disclosed | US | 2017 | <https://patents.google.com/patent/US9593168B2/en>